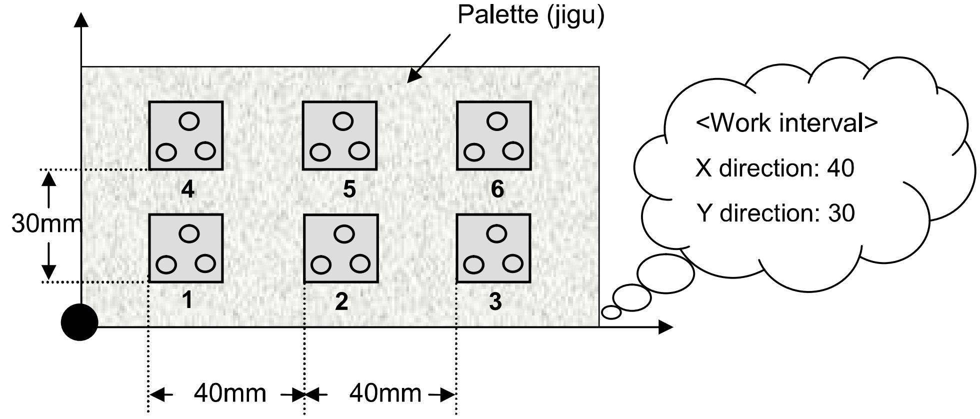
11-1 Auto run(serial measurement of multiple units)

The works of the same type are mounted on the palette (jigu) in a regular interval so that all the works can be measured by one operation. An example of measurement in which the works are fixed on the palette is shown below.



(1) A plan for specifying the reference of palette is prepared.

* The measurement plan name is assumed to be "jigu."

1. Open "Create New Measurement Plan" and specify a base alignment at the front left angle of the palette.
2. Save and close the measurement plan (It is not necessary to specify a clearance plane or a Characteristic).

(2) Prepare a plan of the work to measure.

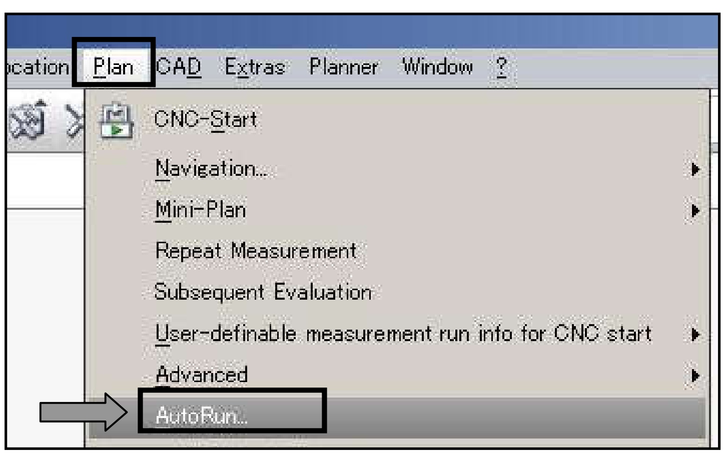
1. Open "Create New Measurement Plan" and prepare a measurement work plan for the work 1. The measurement plan name is assumed to be "test_a" here. Also, specify the clearance plane and define the Characteristic.
2. After the measurement plan is prepared, once execute the measurement plan to check the operation and the output of the result.
3. Save and close the measurement plan.

(3) Next, specify the Auto Run setup options.

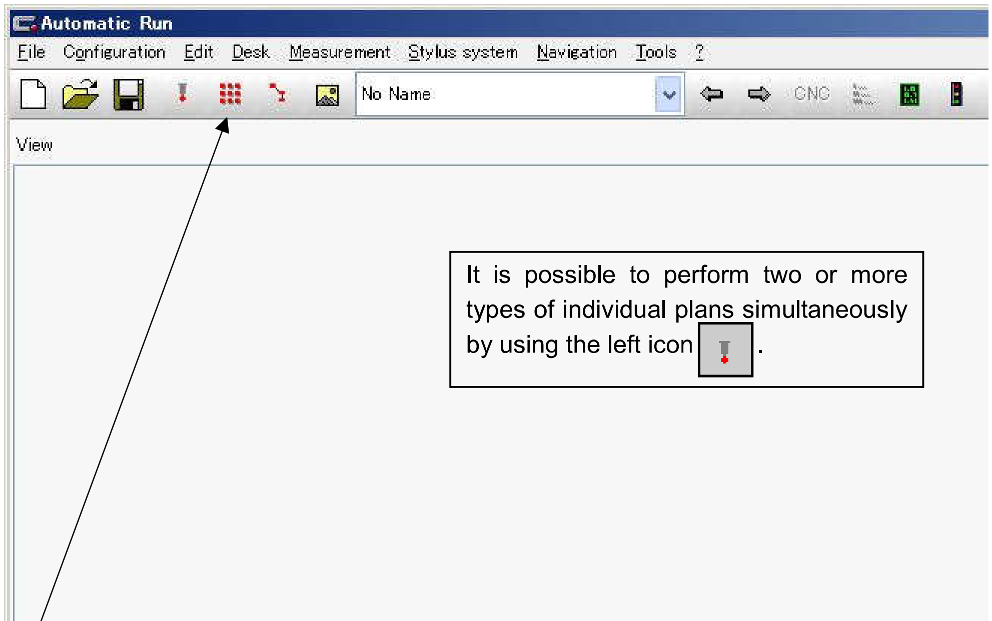
If there is a measurement plan opened, close all of them.

It is not possible to specify the Auto Run setup when another Measurement Plan is kept open.

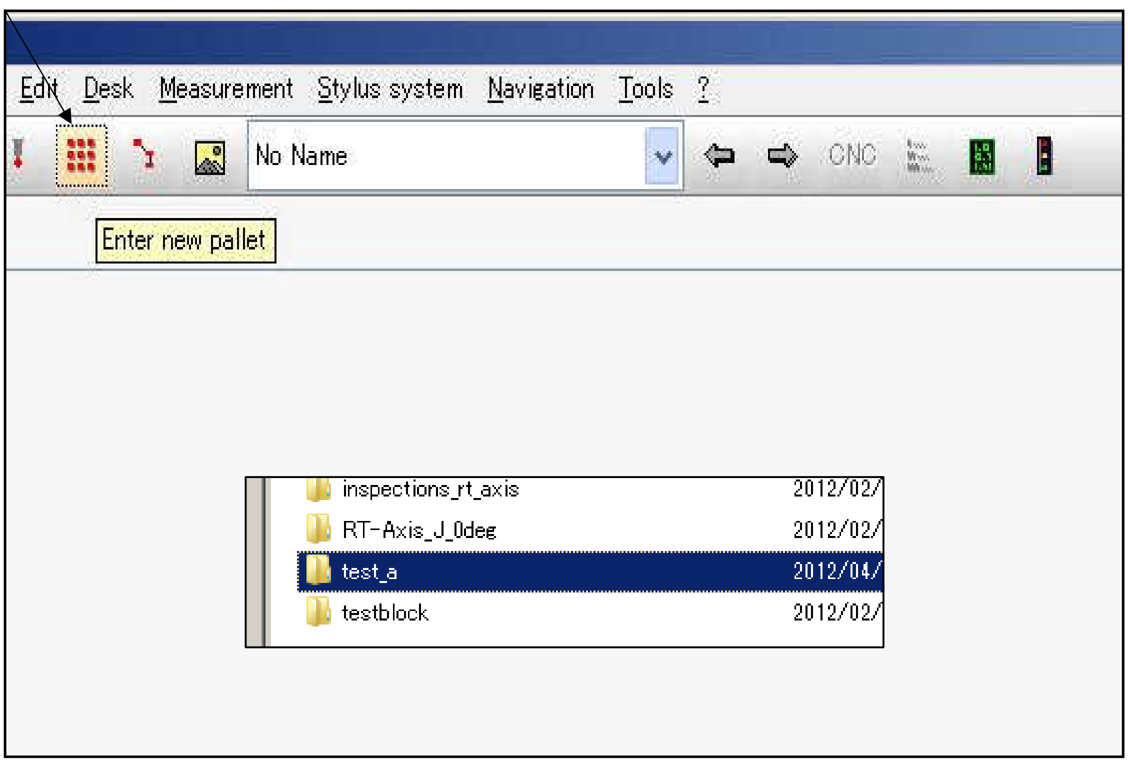
1. Select "**Plan**" and "**Auto Run**."



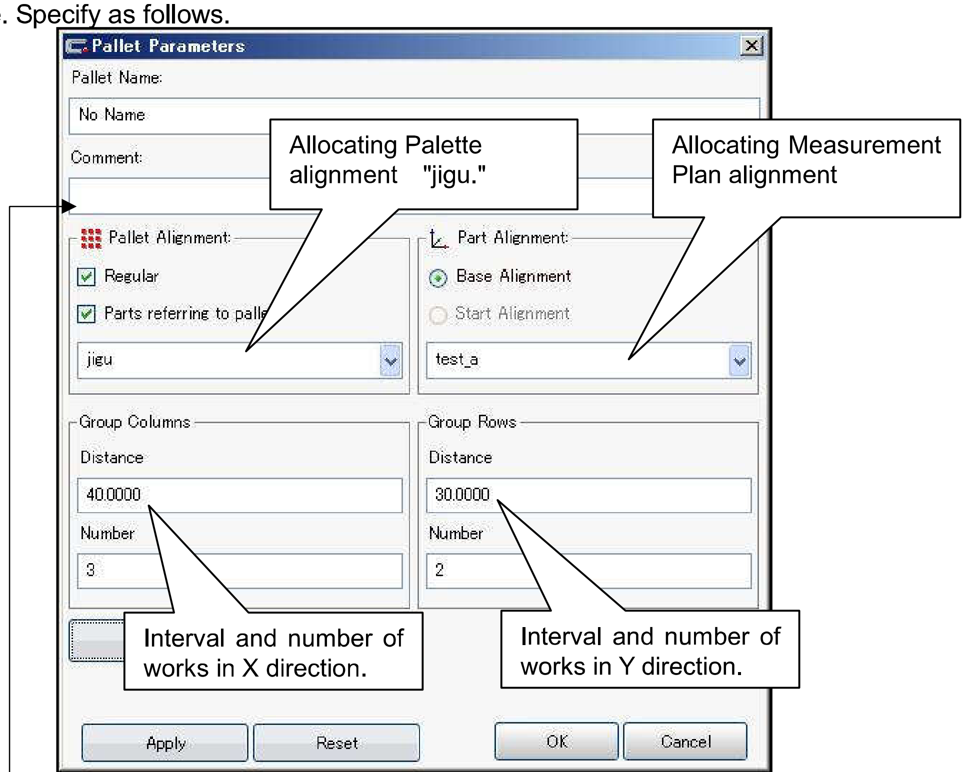
2. Automatic Run window is displayed.



3. File Open window is displayed by selecting **"Palette" icon**. In the window, click the Measurement Plan "test_a" and "Open", then click "inspection" and "Open."

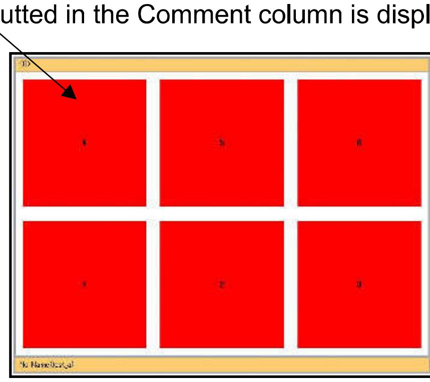


4. Pallet Parameter screen is displayed by opening the Measurement Plan on palette reference. Specify as follows.

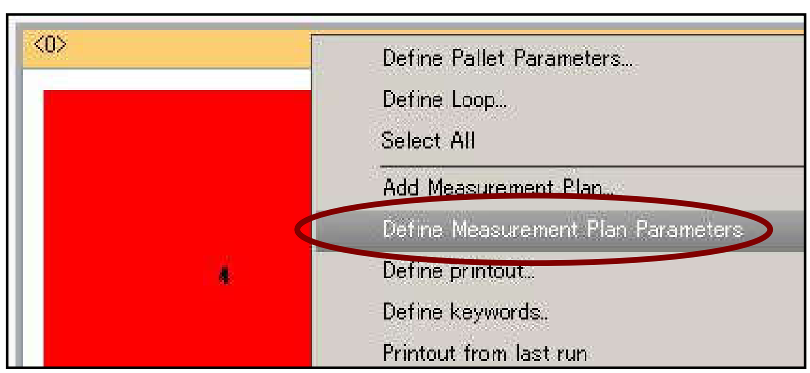


Comments can be input in the Comment column. When the setup is finished, click "OK" to close the window.

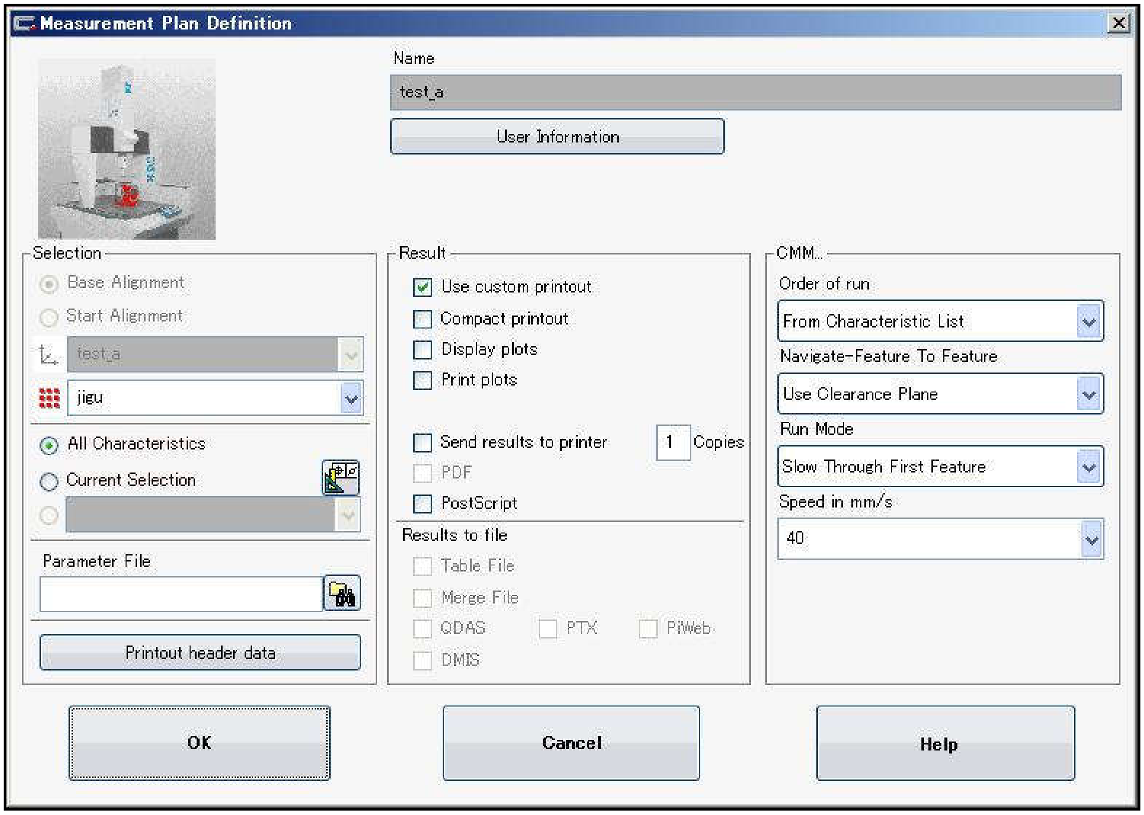
5. A palette is created on the screen.
(The comment inputted in the Comment column is displayed on this part)



6. Select again the pallet by the left click and right click the mouse to select the [Define Measurement plan parameters].



7. Measurement Plan Definition screen is displayed. Specify each condition and click "OK."
   No measurement plan is executed here; only the condition is specified.



8. Then, the setup for Auto Run is completed.
   Select "**File**" and "**Save As**" and input a file name.
   (The file is saved as "File Name. arn")

11-2 Start of Measurement by Auto Run

Before performing measurement by Auto Run!

Since the position on the measurement machine to which the jigs are fixed may be changed if the jigs are not fixed all the time, it is necessary to teach the jig (palette) position and the 1st work position.

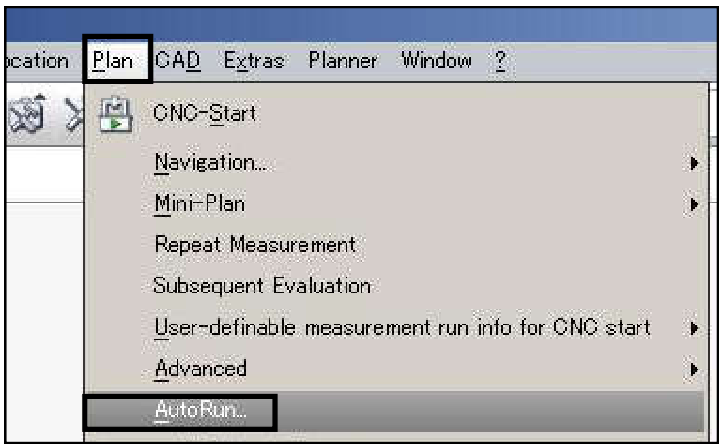
So, before starting Auto Run measurement:

・Open the Palette Measurement Plan, update the Base Alignment, save it, and close the window.

・Open the Measurement Plan to execute, update only the Base Alignment of the 1st work, save it, and close the window.

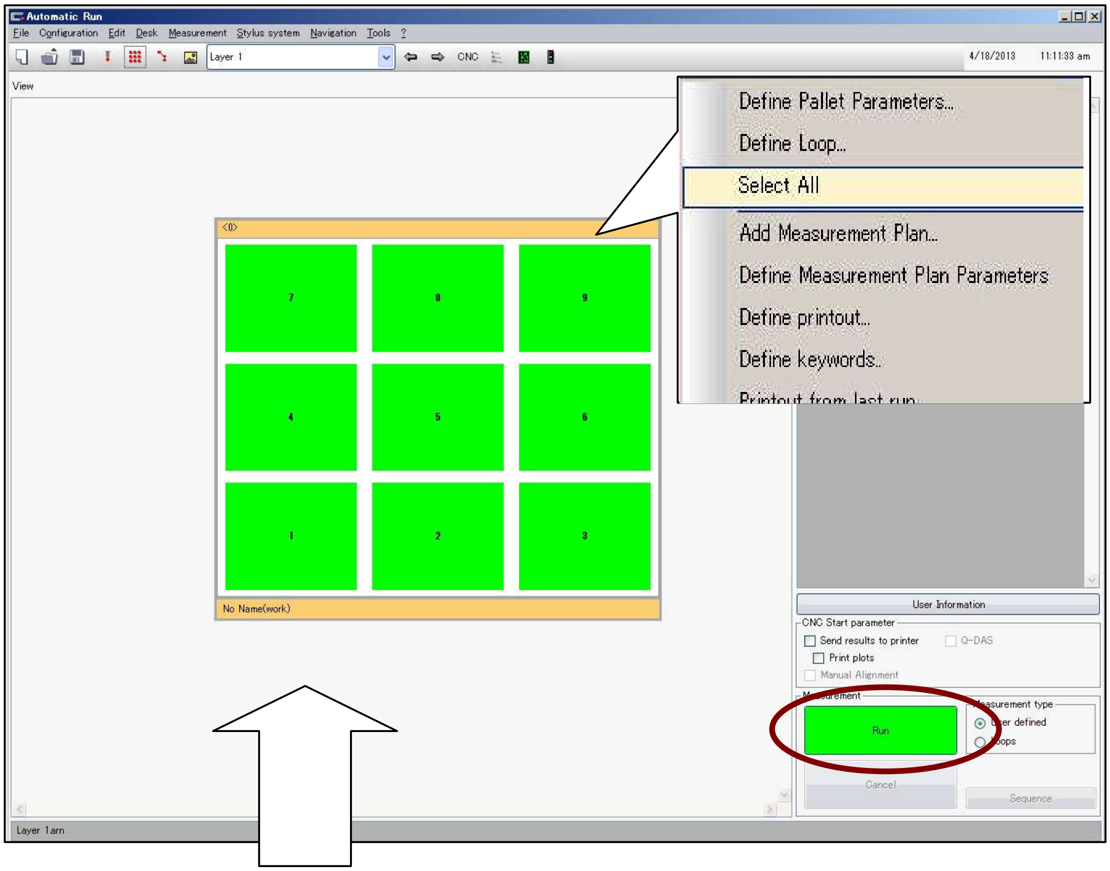
(1) Open the Auto Run Measurement Plan.

Select **Plan and Auto Run**.



(2) Open the file. Select **File and Open** to select the file name and click "Open."

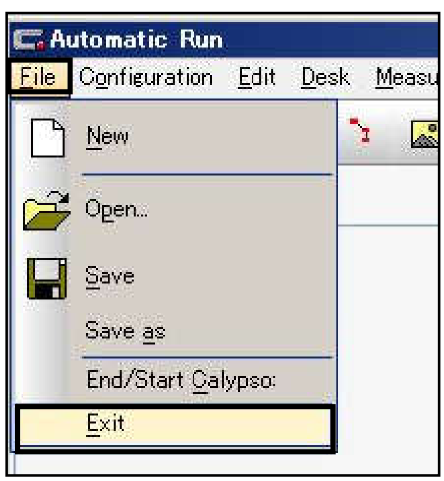
(3) Palettes are displayed. Right-click to select "**Select All**" to turn the palettes in green, then click "Run." The measurement is started.



Reference!

If the number of works is few, select only the palettes to be measured and turn them in green to perform serial measurement of the number of works required. The palettes in red are not measured.

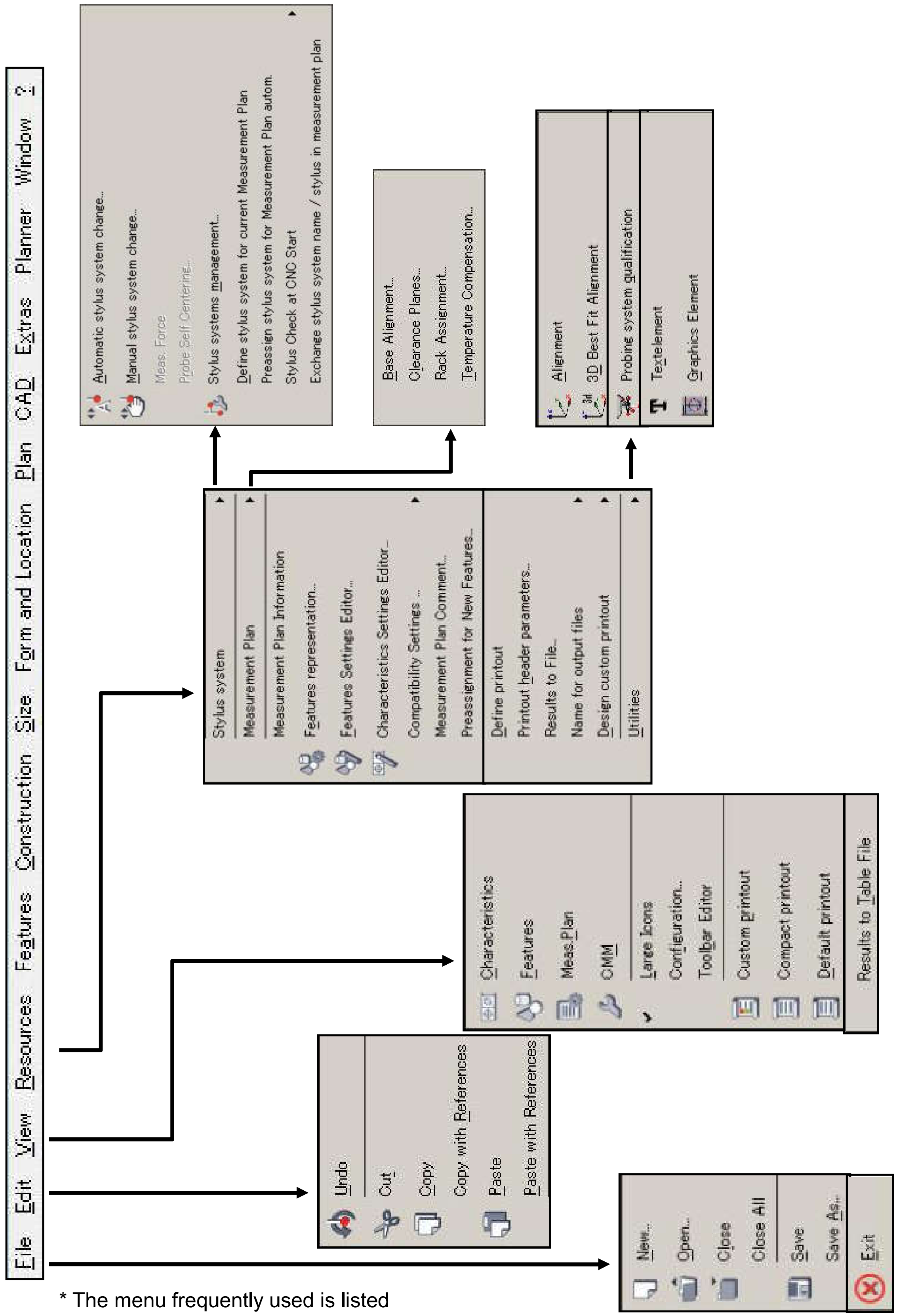
(4) When the measurement is finished, select "**File**" and "**Exit**."



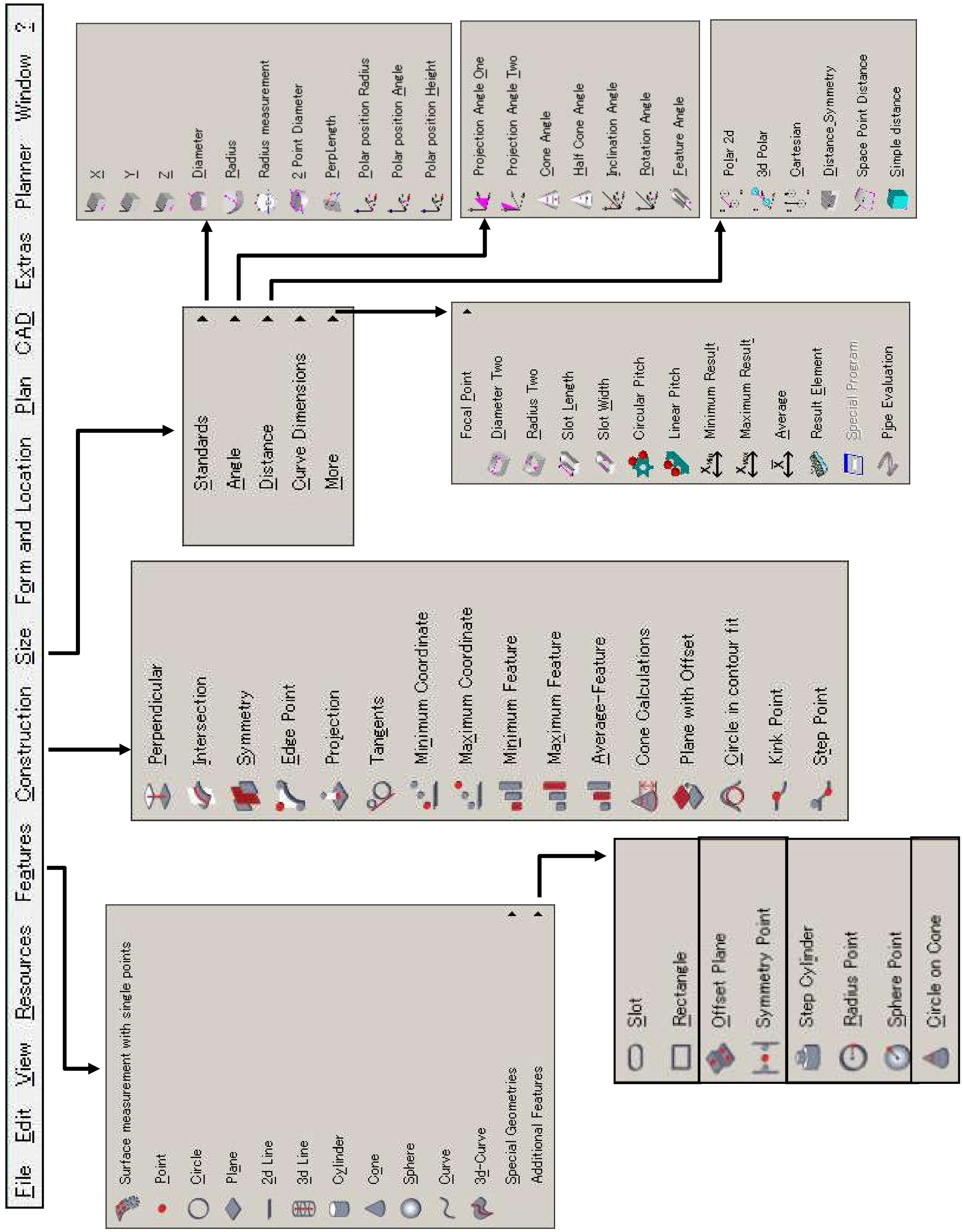
《 Memo 》

Appendix

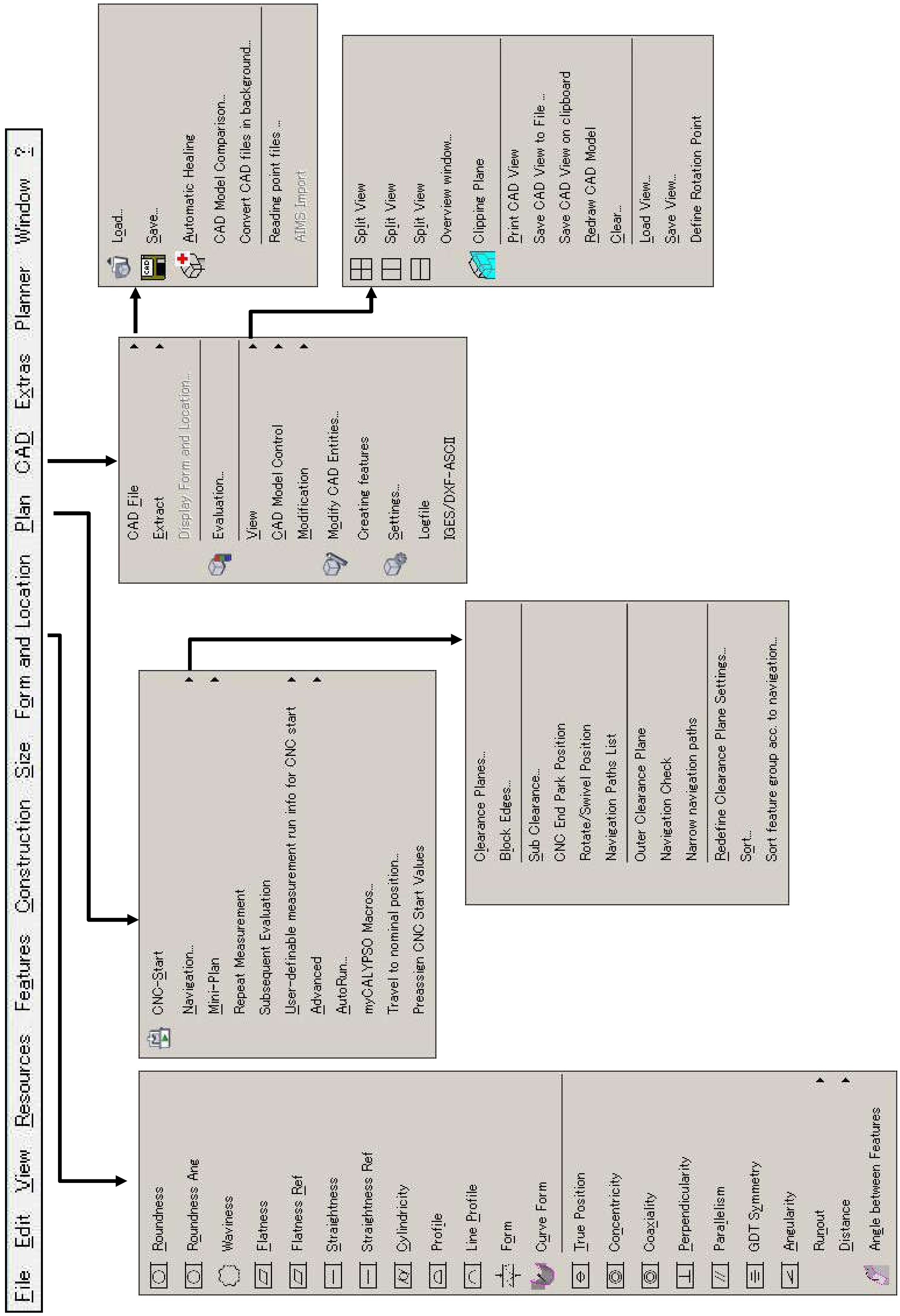
Appendix1. Menu bar list



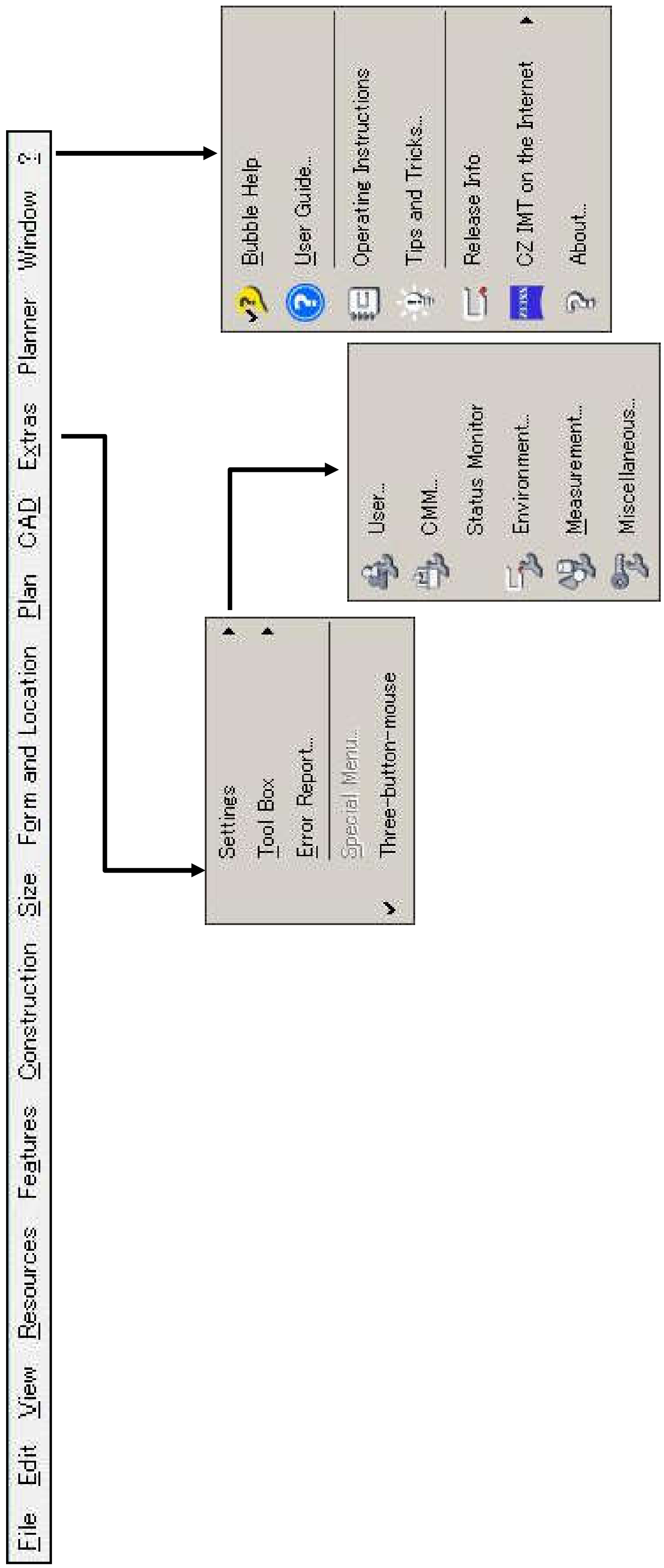
* The menu frequently used is listed



* The menu frequently used is listed

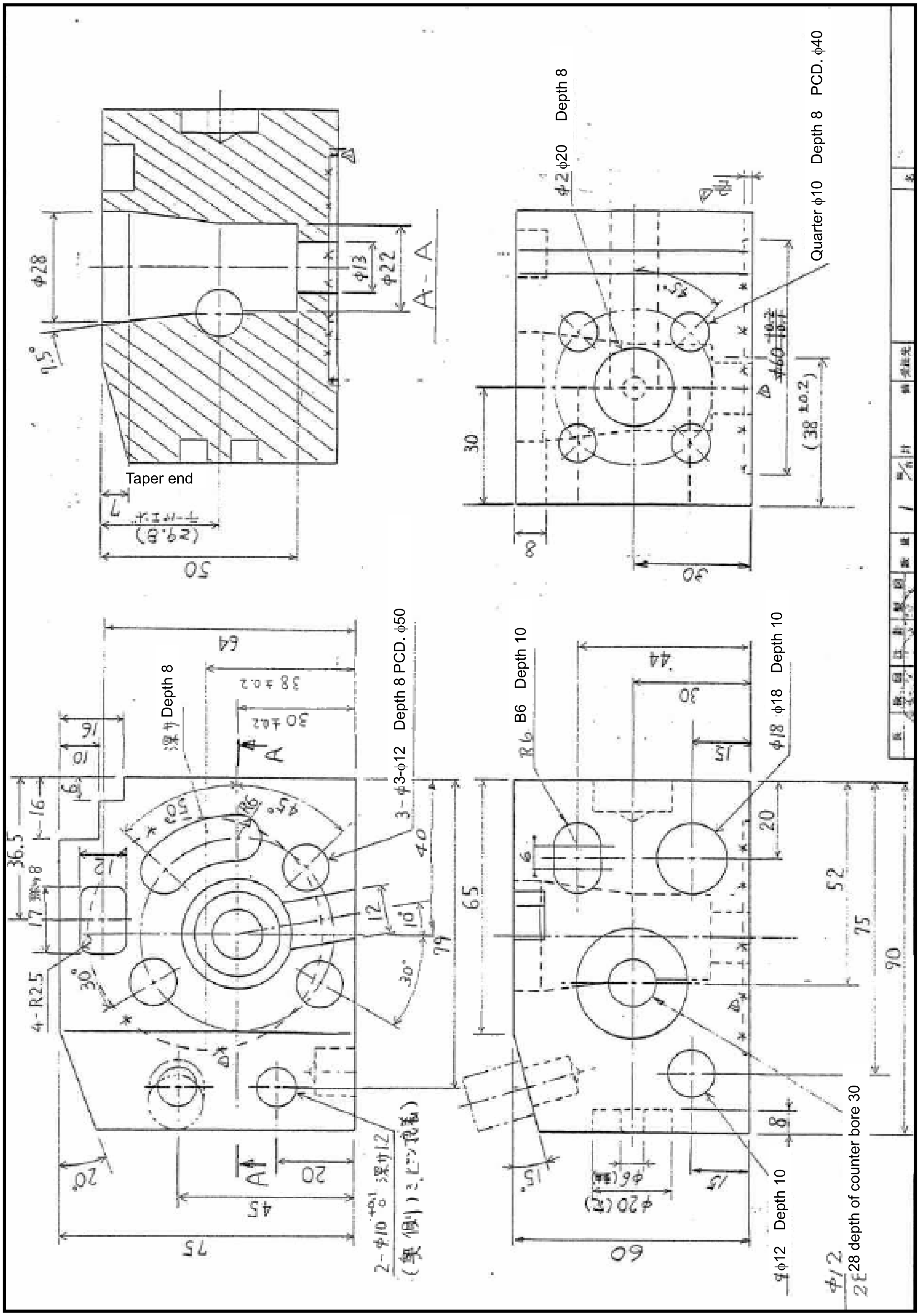


* The menu frequently used is listed



* The menu frequently used is listed

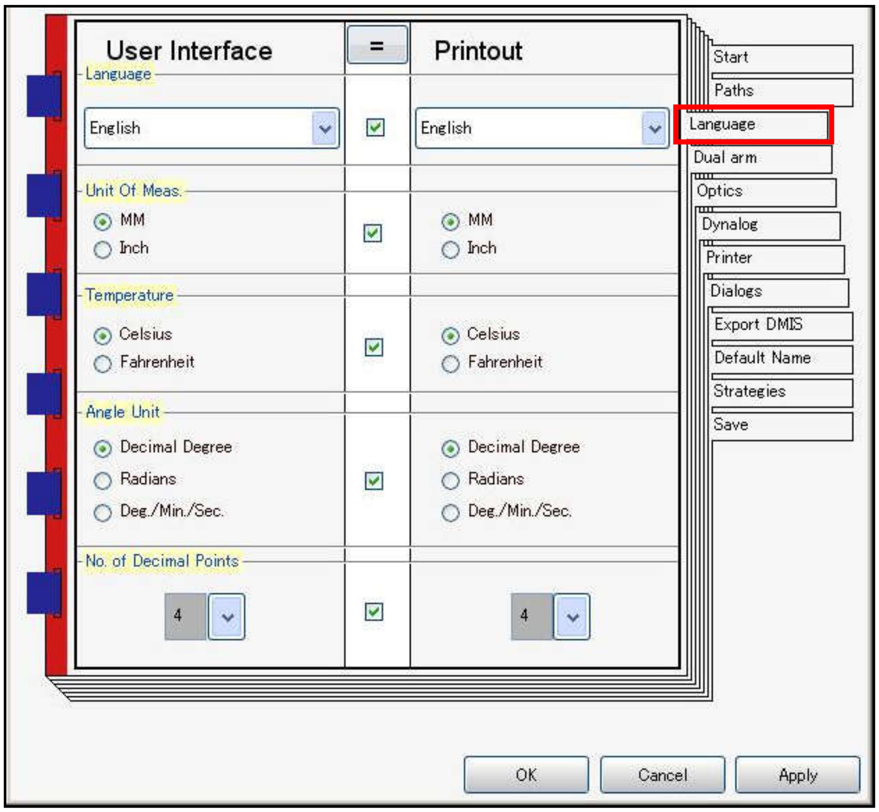
Appendix2. Test block drawing



Appendix3. How to change the unit

1. From the menu, select Extras, Workroom, and Environment.

2. The following screen is displayed. Click "Language" and select the option in each column. Then, click "Apply" and "OK" to close the window.
   It is recommended to specify these options at the start of measurement.



* Notes for converting the unit of length into inches

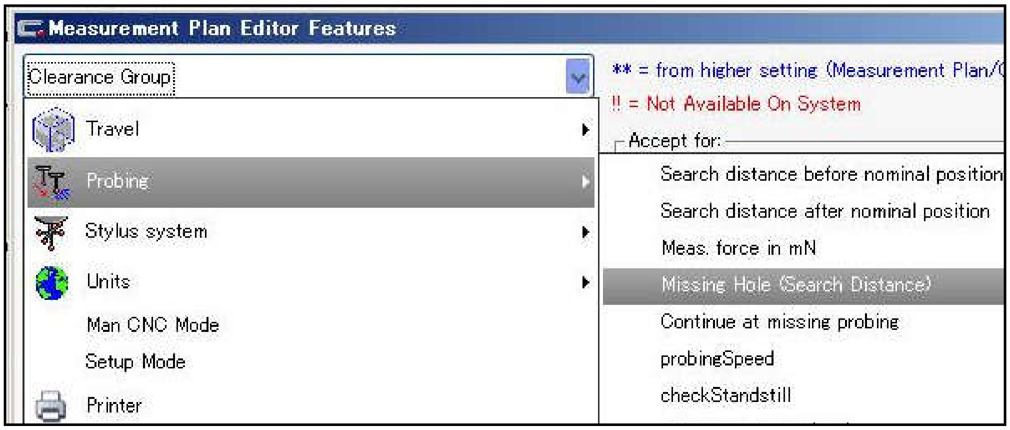
To specify the inch as the unit of length, change the setup of the "round off of design value" too.
After changing the unit, select Extras, Workroom, and Measurement .
Click "Nominal" tab and change the value to 0.001 for the Nominal Default Round off.

Add two more digits than the case with using the "mm" in order to equalize the round off outcome.
If the round off setup is inappropriate, the position of CNC measurement may be shifted greatly.
For that reason, never fail to follow the setup details.

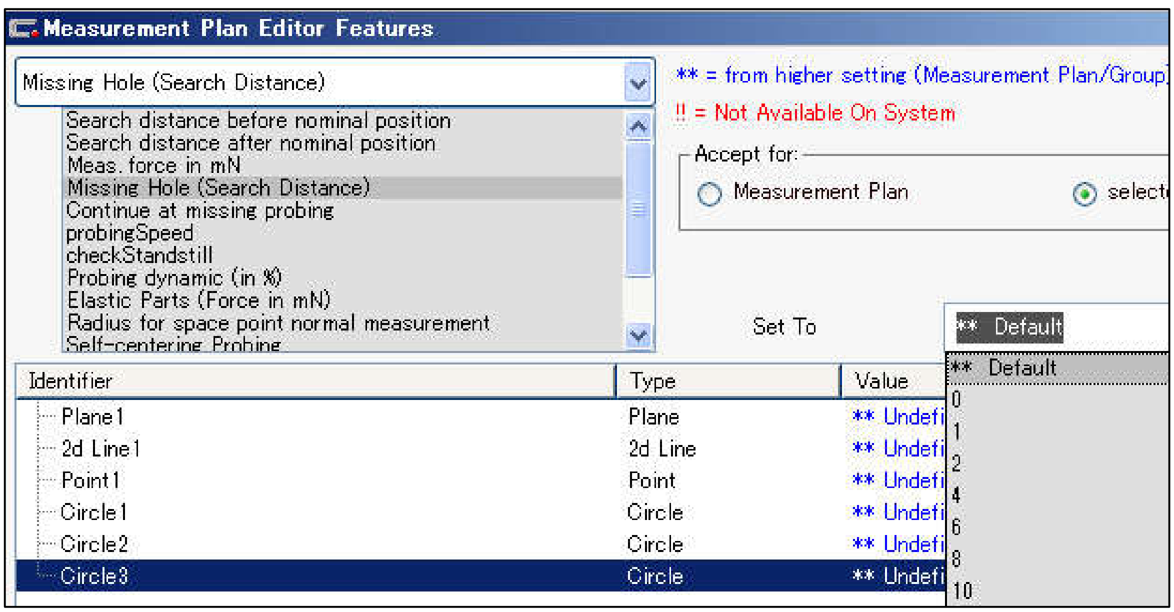
Appendix4. Missing Hole function

If a work is not finished completely and there is no hole which was specified on the plan, the work is supposed to be stopped by collision. However, this function enables CNC measurement performing to the last without interruption by collision.

1. Open the Measurement Plan Editor Features, select the "Probing" and "Missing Hole (Search Distance)",



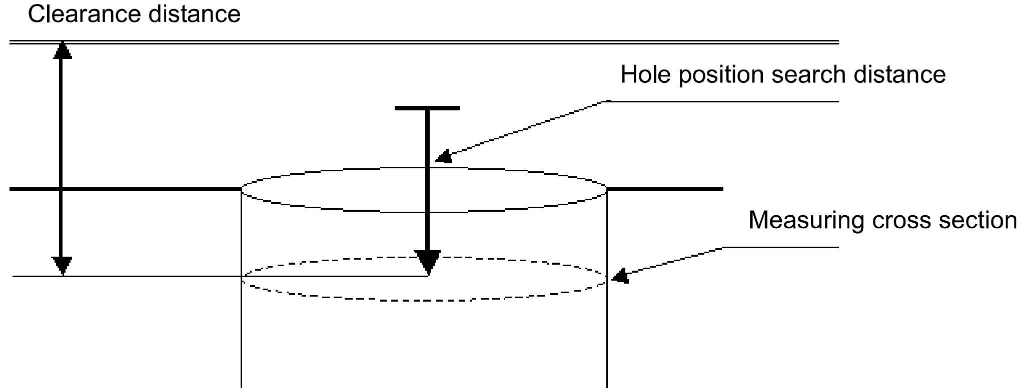
2. Select the hole feature to apply the Missing Hole (Search Distance) function, and specify the Search Distance from the "Set To."



* The setup of the Hole Position Search Distance is reflected on the measurement as follows.

Search start: The position that moves up for the amount of the search distance from the measuring cross section.

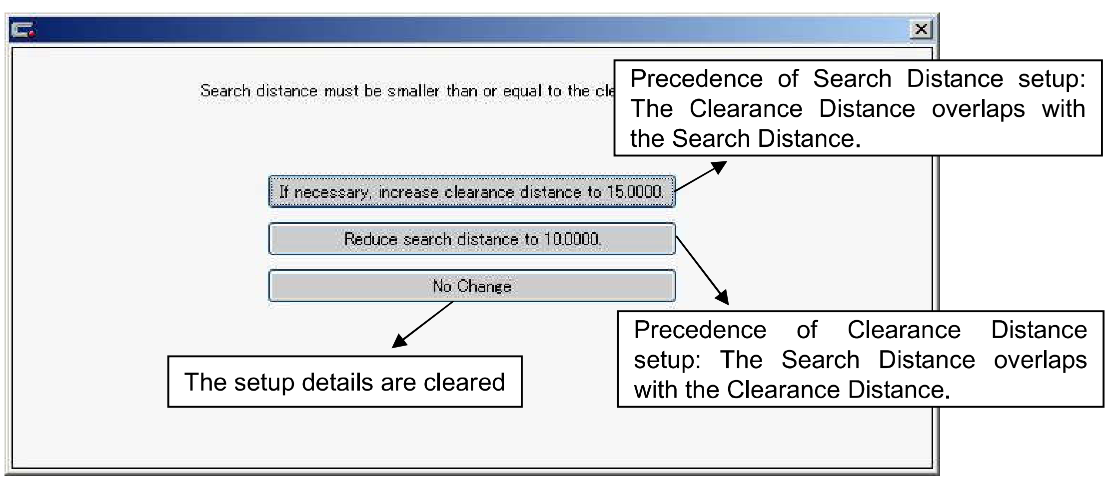
Search end: Height of the measuring cross section



Note - Search distance and clearance distance

For specifying the Search Distance of the Missing Hole, do not put a value greater than the Clearance Distance.

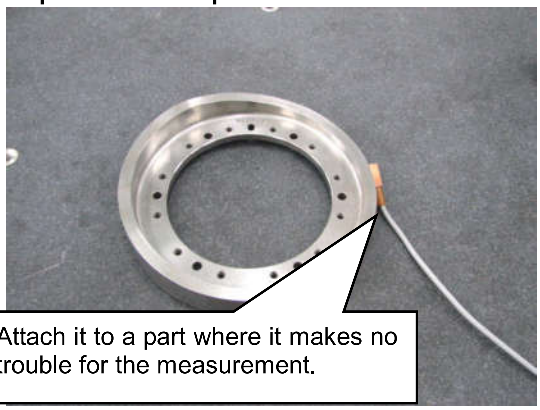
If a greater value is input, the following error message is displayed.

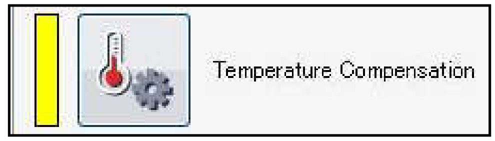


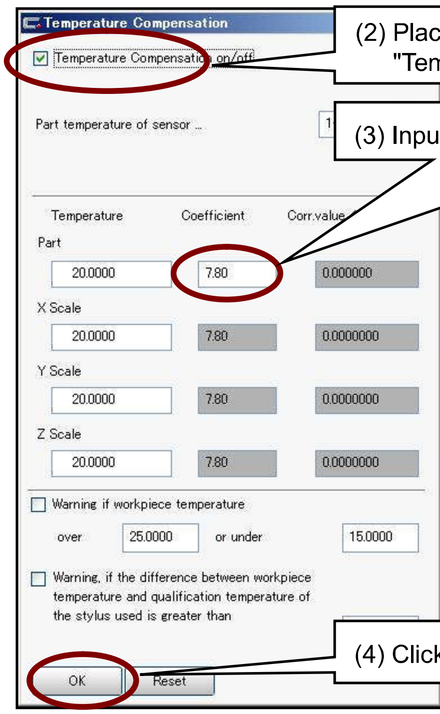
Appendix5. Temperature-Compensation function

For the scale and work of each axis, Calypso can consider dimensional change due to the temperature condition at the time of measurement. By effecting the temperature-correction, it is possible to obtain the result corrected to the value measured at 20 degrees.

(1) Before the measurement, mount the temperature sensor on the work,and click **"Temperature-Compensation" icon** of the Prerequisites List.







The reference values of some materials are shown below:

<Material>	<Thermal expansion coefficient>
Aluminum	23. 1
Copper	16. 5
Magnesium	24. 8
Nickel	13. 4
Carbon steel	10. 7
Titanium	8. 6

Reference:"Science Chronology" compiled by National Astronomical Observatory, Issued by Maruzen

* The values to input in the column are as follows:

The expansion coefficient corresponding to the material of the work.

It is corrected allowing for the expansion of the scale and the work for the measurement machine.

0

It is corrected allowing only for the expansion of the scale for the measurement machine.

Then, the temperature-compensation setup is completed.

Continue the measurement.

It is corrected based on the temperature at the time of the Alignment measurement just after the start of the automatic measurement following to the measurement plan completion.

Appendix6. Calibration of disk stylus

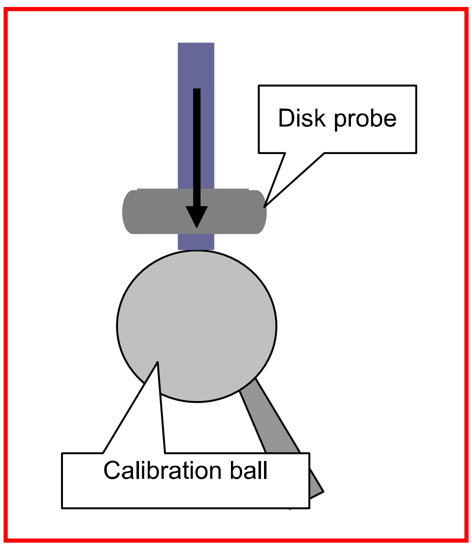
Disk probe can be calibrated only manually.

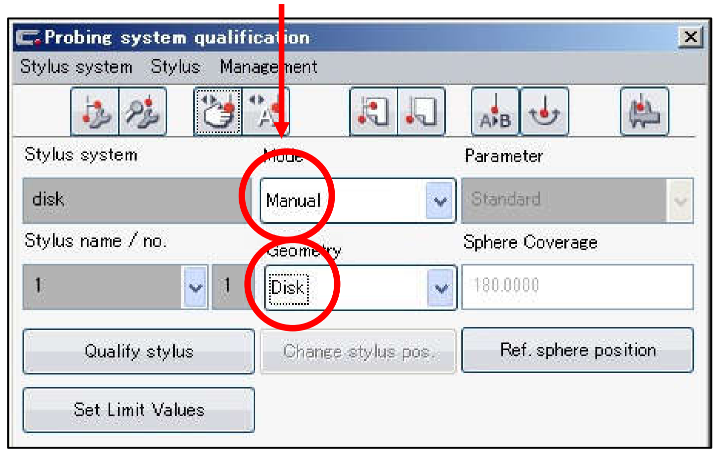
An automatic calibration is possible: for the caliper mark icon by the same procedure of manual calibration; and with the measurement plan for stylus calibration

Procedure 1: Register the position of the Reference sphere with the Master Probe.
Procedure 2: Attach the disk stylus.
Procedure 3. Change the "mode" as Manual and "Geometry" as Disk.
Procedure 4: Perform stylus calibration

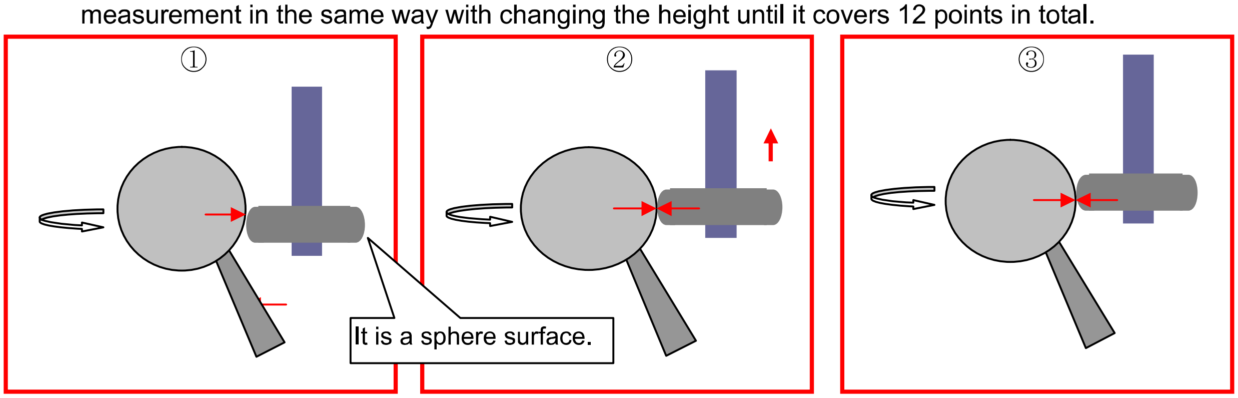
At first, perform manual calibration.

The first one point is not a measurement point but it teaches stylus axis direction.





The second and subsequent points become the measurement points which are captured as calibration data. At the beginning, set the height so that the upper part of the sphere surface contacts with the Reference Sphere. With the height, four points on the circumference are measured. Perform the measurement in the same way with changing the height until it covers 12 points in total.

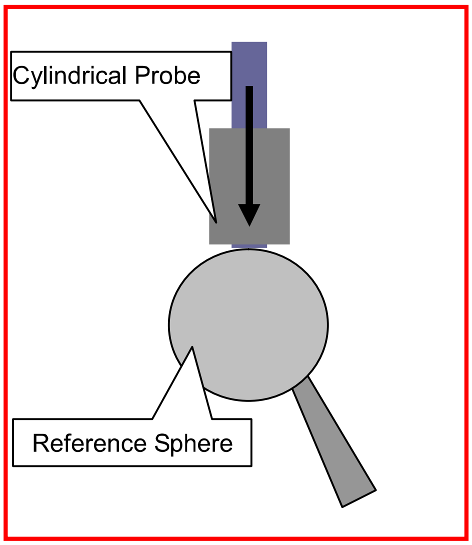


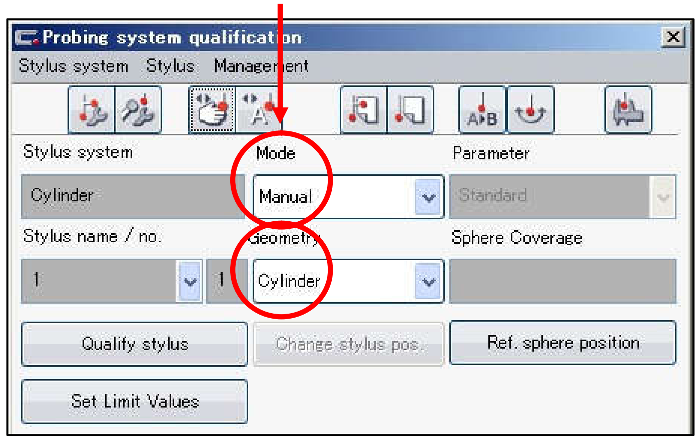
Although it is desirable to perform measurement over a wide range as possible, take care not to let the edge collide with the Reference Sphere.

Appendix7. Calibration of cylindrical stylus

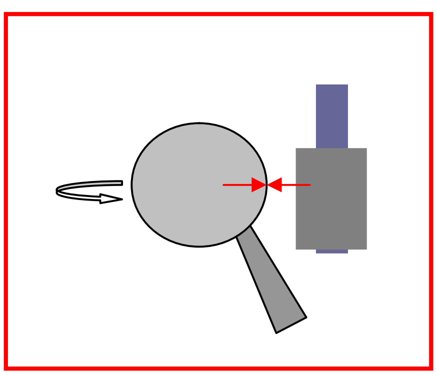
Only the manual calibration is applicable to the Cylindrical stylus as same as the disk probe.
Auto calibration is applicable in the order implemented by manual calibration by the Slide calipers icon and/or Stylus calibration measurement plan.

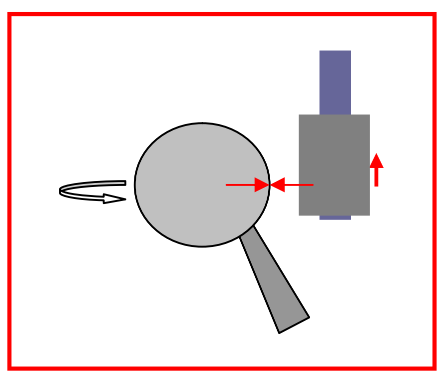
① Define the position of the Reference Sphere by the Master Probe.
② Mount the cylindrical stylus.
③ Set the [Mode] to Manual and [Geometry] to Cylinder.
④ Execute the calibration of the stylus.
Implement the calibration by manual at first.
First point is the one to show the axis direction of the stylus but not the measurement point.





Points after the second point shall become the measurement points to be adopted as the calibration data.
First, align the cylindrical stylus to the height that a little lower part of the measuring height contacts the calibration ball.
Measure three points at that height.
Next, move the cylindrical stylus a little higher that the measuring height and implement measurement in the same manner total six points. And more points as needed.





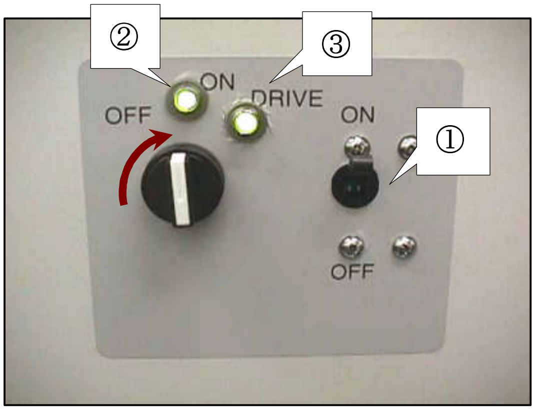
Appendix8. Startup of old control panel

Startup of measurement machine main unit

(1) Turn on the main power of the controller.

(2) Turn on the controller switch.

〈Controller〉



(3) After about 30 seconds, the display on the control panel turns into the Stylus Information from "ZEISS." Turn on the "DRIVE."

(4) Turn on the printer power

(5) Turn on the monitor power

(6) Turn on the PC power